Case Study: Monitoring & Leak Detection in a Water Network

From Unstructured Raw Sensor Telemetry to a Relational Data Model

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1. Background and Scenario

The municipal utility **Bergisches Land Waterworks (BLWW)** operates a branched and looped drinking water distribution network fed by a central elevated reservoir (`HB-North`) that supplies three demand zones: **Residential Area West** (`V-West`), **Industrial Park South** (`V-South`), and **Hospital Complex East** (`V-East`).

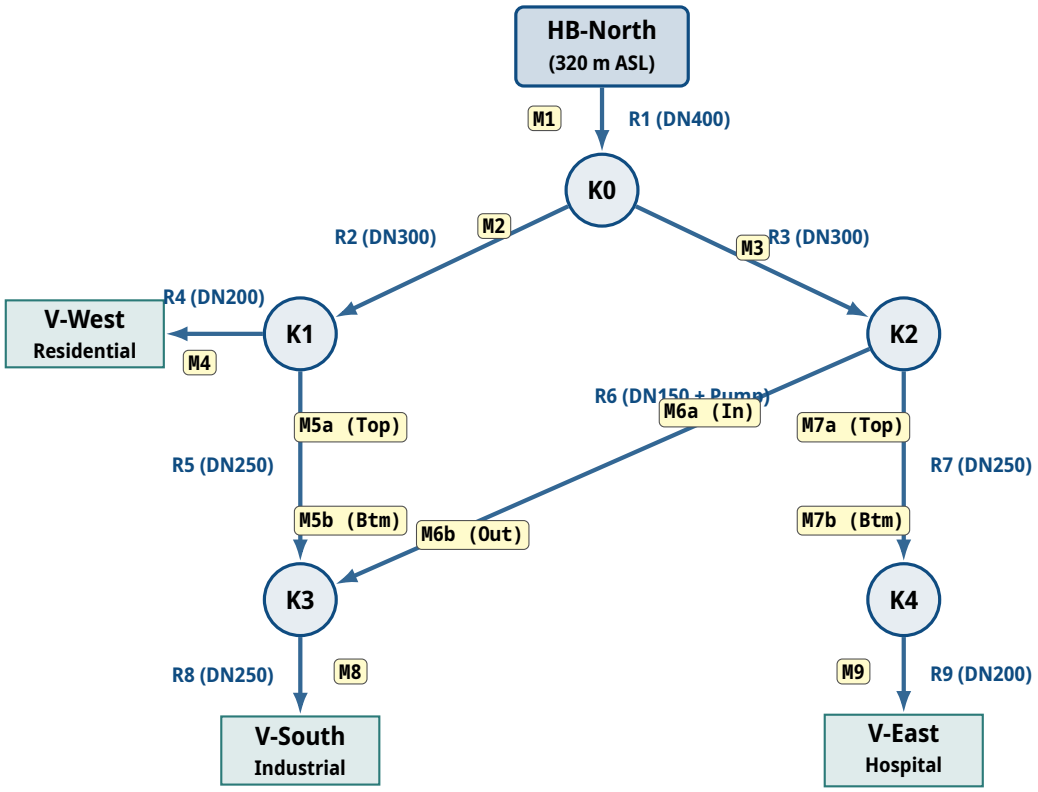
To ensure supply reliability during maintenance, the network is designed not as a simple tree, but as a **meshed network (looped topology)** featuring a cross-connection and a booster pump. Along the pipelines, IoT sensor stations are installed at strategic points, writing telemetry data (volumetric flow rate, flow velocity, water pressure, temperature) along with static asset metadata into a central landing zone (raw CSV dump) at 15-minute intervals.

The Challenge:

1. The current data storage is completely denormalized. With every single sensor reading, all pipe, node, and sensor metadata are stored redundantly. Furthermore, some legacy sensors report flow in m^3/h, newer sensors report in liters per second (l/s), and for certain sensors the volumetric flow unit failed, leaving only the flow velocity v (m/s).
2. Since last night, the network control center has reported a **significant, continuous water loss** across the overall system. No visible pipe burst has surfaced on the streets (indicating an underground seepage into the soil or sewer system).

2. Network Topology and Sensor Layout

The diagram below illustrates the elevated reservoir, the five junction nodes (`K0` to `K4`), the nine pipeline segments (`R1` to `R9`), and the□□ installed flow and pressure measurement stations (`M1` to `M9`).



Technical Specifications of Network Components

- **Junction Nodes (Chambers):** Pipelines branch or merge at nodes K0 to K4. In an intact, leak-free node, Kirchhoff's first law (continuity equation for incompressible fluids) holds: $\sum \dot{V}_{\text{in}} = \sum \dot{V}_{\text{out}}$.
- **Pipe Segments (R1–R9):** Every pipe has a specific internal diameter D (nominal diameter DN in mm, e.g., DN300$D = 300\,\text{mm} = 0.3\,\text{m}$), a material (Cast Iron, Ductile Iron, HDPE, Steel), a construction year, and a total length L.
- **Measurement Stations (M1–M9):** Several long trunk mains (R5, R6, R7) are equipped with sensors at both the upstream start (a) and downstream end (b) of the pipe, whereas shorter pipes (R1, R2, R3, R4, R8, R9) have only a single sensor.

3. The Raw Dataset (Excerpt from IoT Gateway Dump)

The table below shows an excerpt from the raw data export (`raw_sensor_dump.csv`) for two timestamps: **02:00** (nighttime low-demand period) and **08:00** (morning peak demand).

Important Technical Notes on the Telemetry Data:

- **Mixed Units:** Older ultrasonic sensors report the volumetric flow rate Q in m³/h, while newer electromagnetic flow meters (Magmeters) report Q in liters per second (l/s). Conversion factor: $1\,\text{l/s} = 3.6\,\text{m}^3/\text{h}$.
- **Missing Flow Rate Values (`NULL`):** At sensor stations `M3` and `M7a`, the flow integration unit malfunctioned (`NULL`). However, these sensors still reliably measure the mean flow velocity v in m/s. Using the circular pipe cross-sectional area $A = \pi \cdot (D/2)^2$, the volumetric flow rate can be derived via:

$$Q\ [\text{in m}^3/\text{s}] = A \cdot v = \pi \cdot \left(\frac{D}{2}\right)^2 \cdot v \quad \Longleftrightarrow \quad Q\ [\text{in m}^3/\text{h}] = \pi \cdot \left(\frac{D}{2}\right)^2 \cdot v \cdot 3600$$

- **Booster Pump Operation in R6:** Pipe R6 houses an inline booster pump that pumps water from K2 toward K3 (providing a pressure boost of approx. $+1.5$ bar).

Time	M-ID	Pipe	From→To	Pos (m)	Mat. / Year	DN / Len	v (m/s)	Q_val	Unit	p (bar)
02:00	M1	R1	HB→K0	50	Steel / 1998	400mm / 500m	0.442	200.00	m3/h	4.80
02:00	M2	R2	K0→K1	100	CastIron / 1974	300mm / 800m	0.446	113.50	m3/h	4.65
02:00	M3	R3	K0→K2	200	HDPE / 2015	300mm / 600m	0.340	NULL	NULL	4.70
02:00	M4	R4	K1→VW	50	HDPE / 2018	200mm / 400m	0.248	7.80	l/s	4.30
02:00	M5a	R5	K1→K3	20	CastIron / 1974	250mm / 950m	0.410	72.50	m3/h	4.28
02:00	M5b	R5	K1→K3	920	CastIron / 1974	250mm / 950m	0.410	72.50	m3/h	4.10
02:00	M6a	R6	K2→K3	30	Ductile / 2005	150mm / 700m	0.409	7.22	l/s	4.55
02:00	M6b	R6	K2→K3	680	Ductile / 2005	150mm / 700m	0.409	26.00	m3/h	5.85
02:00	M7a	R7	K2→K4	25	CastIron / 1968	250mm / 1100m	0.342	NULL	NULL	4.52
02:00	M7b	R7	K2→K4	1050	CastIron / 1968	250mm / 1100m	0.255	45.00	m3/h	3.60
02:00	M8	R8	K3→VS	40	Ductile / 2010	250mm / 300m	0.557	27.36	l/s	4.05
02:00	M9	R9	K4→VO	30	HDPE / 2019	200mm / 250m	0.398	45.00	m3/h	3.55
08:00	M1	R1	HB→K0	50	Steel / 1998	400mm / 500m	1.105	500.00	m3/h	4.50
08:00	M2	R2	K0→K1	100	CastIron / 1974	300mm / 800m	1.128	287.00	m3/h	4.20
08:00	M3	R3	K0→K2	200	HDPE / 2015	300mm / 600m	0.837	NULL	NULL	4.35
08:00	M4	R4	K1→VW	50	HDPE / 2018	200mm / 400m	1.035	32.50	l/s	3.85
08:00	M5a	R5	K1→K3	20	CastIron / 1974	250mm / 950m	0.905	160.00	m3/h	3.82
08:00	M5b	R5	K1→K3	920	CastIron / 1974	250mm / 950m	0.905	160.00	m3/h	3.45
08:00	M6a	R6	K2→K3	30	Ductile / 2005	150mm / 700m	0.865	15.28	l/s	4.15
08:00	M6b	R6	K2→K3	680	Ductile / 2005	150mm / 700m	0.865	55.00	m3/h	5.30
08:00	M7a	R7	K2→K4	25	CastIron / 1968	250mm / 1100m	0.894	NULL	NULL	4.12
08:00	M7b	R7	K2→K4	1050	CastIron / 1968	250mm / 1100m	0.818	144.50	m3/h	3.15
08:00	M8	R8	K3→VS	40	Ductile / 2010	250mm / 300m	1.217	59.72	l/s	3.40
08:00	M9	R9	K4→VO	30	HDPE / 2019	200mm / 250m	1.278	144.50	m3/h	3.08

Additional Asset Metadata from Enterprise Asset Management (to be included in the schema):

- Every node (HB, K0–K4, VW, VS, VO) has GPS coordinates (`lat`, `lon`), an elevation in meters above sea level (ASL), and a node type (*Reservoir, Junction, DemandZone*).
- Every measurement station (M1–M9) has a hardware serial number, a sensor technology type (*Ultrasonic, Magmeter*), a last calibration date, and its relative installation position (`Pos (m)`, measured from the upstream start node of the respective pipe).

4. Student Tasks

Part A: Database Design & Normalization (Data Modeling)

1. **Anomaly Analysis:** Using concrete examples from the raw dataset above, explain what redundancies and database anomalies (update, insertion, and deletion anomalies) will occur if this flat table structure is used directly in production.
2. **Relational Modeling (3NF):** Design a clean relational database schema in **Third Normal Form (3NF)** that captures the network topology (nodes, directed pipe segments), sensor metadata (including exact installation position along the pipe), and high-frequency time-series telemetry (measurements).
3. **DDL Implementation:** Write the corresponding SQL `CREATE TABLE` statements including appropriate data types, primary keys (`PRIMARY KEY`), foreign keys (`FOREIGN KEY`), and domain constraints.

Part B: ETL Pipeline & Data Cleansing (Data Transformation)

4. **Standardization & Imputation Logic:** Formulate the transformation logic (as an SQL query or Python/Pandas script) that normalizes all volumetric flow rates to the standard unit m^3/h and automatically imputes missing `Q_val` entries (`NULL`) from the flow velocity v and nominal pipe diameter D.
5. **Calculation of Missing & Converted Values:** Calculate the exact volumetric flow rates in m^3/h (rounded to one decimal place) for the missing sensors `M3` and `M7a`, as well as the unit-converted flow rates for `M4`, `M6a`, and `M8` at both timestamps (02:00 and 08:00).

Part C: Forensic Data Analysis – Where is the Network Losing Water?

Statement from the Control Center: *"Our overall water balance indicates that the network is continuously losing water. There is at least one undetected leak in the system—possibly even more!"*

6. **Node and Pipe Segment Mass Balance:** Calculate the mass balance (inflow vs. outflow in m^3/h) for every pipe segment and every junction node at both timestamps (02:00 and 08:00).
7. **Leak Localization and Physical Explanation:** Pinpoint **where** (on which pipes or between which sensors/nodes) the network is losing water, quantify the water loss at 02:00 and 08:00, and explain why the leakage rate changes counter-intuitively between 02:00 and 08:00 (Hint: examine the pressure readings `p (bar)` and the pipe materials/ages!).